

CryptX

For Encrypting/Decrypting Texts

By: Abdalrahman Ehab Mahmoud
Zeyad Sameh Abdelhady
Roaa Hesham Fathy
Seif Ashraf Abdelaziz
Hussien Mohamed Hussien
Abdelrahman Salah El-Deen

How the Project Works

1. Purpose of the Project:

- This project is a web-based encryption and decryption tool called CryptX.
- It allows users to encrypt plaintext into ciphertext and decrypt ciphertext back into plaintext using cryptographic algorithms.

2. Encryption Process:

- The user enters plaintext in the input field.
- They select a key generation method: LCG (Linear Congruential Generator), RSA, or a Custom Key.
- The backend generates a key (if necessary) and encrypts the plaintext using the CTR Mode (Counter Mode) algorithm.
- The encrypted text (ciphertext) and the encryption key are displayed to the user.

3. Decryption Process:

- The user enters the ciphertext and the decryption key.
- The backend decrypts the ciphertext using the provided key.
- The original plaintext is displayed to the user.

4. Key Generation Methods:

- LCG: Generates pseudo-random keys using a mathematical formula.
- RSA: Uses the RSA algorithm to generate secure encryption keys.
- Custom Key: Allows users to input their own encryption key.

5. Language Support:

- The tool supports multiple languages, including Arabic, using UTF-8 encoding.

Tools and Technologies Used

1. Backend Development:

- Python: The main programming language used for backend logic.
- Flask: A lightweight web framework to handle HTTP requests and serve the application.

2. Frontend Development:

- HTML: For structuring the web pages.
- CSS: For styling the user interface.
- JavaScript: For adding interactivity to the frontend.

3. Encryption Algorithms:

- CTR Mode: A block cipher mode of operation used for encryption and decryption.
- LCG: A simple random number generator for key generation.
- RSA: A widely used public-key cryptosystem for secure key generation.

4. Other Tools:

- UTF-8 Encoding: Ensures compatibility with multilingual text, including Arabic.
- PyInstaller: Used to convert the Python project into a standalone executable file (optional).

How to Run the Project

1. Install Python:

- Download and install Python from the official website: <https://www.python.org/downloads/>.
- Ensure you have Python 3.10 or higher installed.

2. Set Up the Project:

- Clone or download the project files to your computer.

3. Install Dependencies:

- Open a terminal and navigate to the project folder.
- Install the required libraries using:

pip install flask

4. Run the Application:

- Start the Flask server by running:

python app.py

- Open a web browser and go to **http://127.0.0.1:5000** to access the application.

5. Use the Application:

- Use the encryption panel to encrypt text.
- Use the decryption panel to decrypt text.

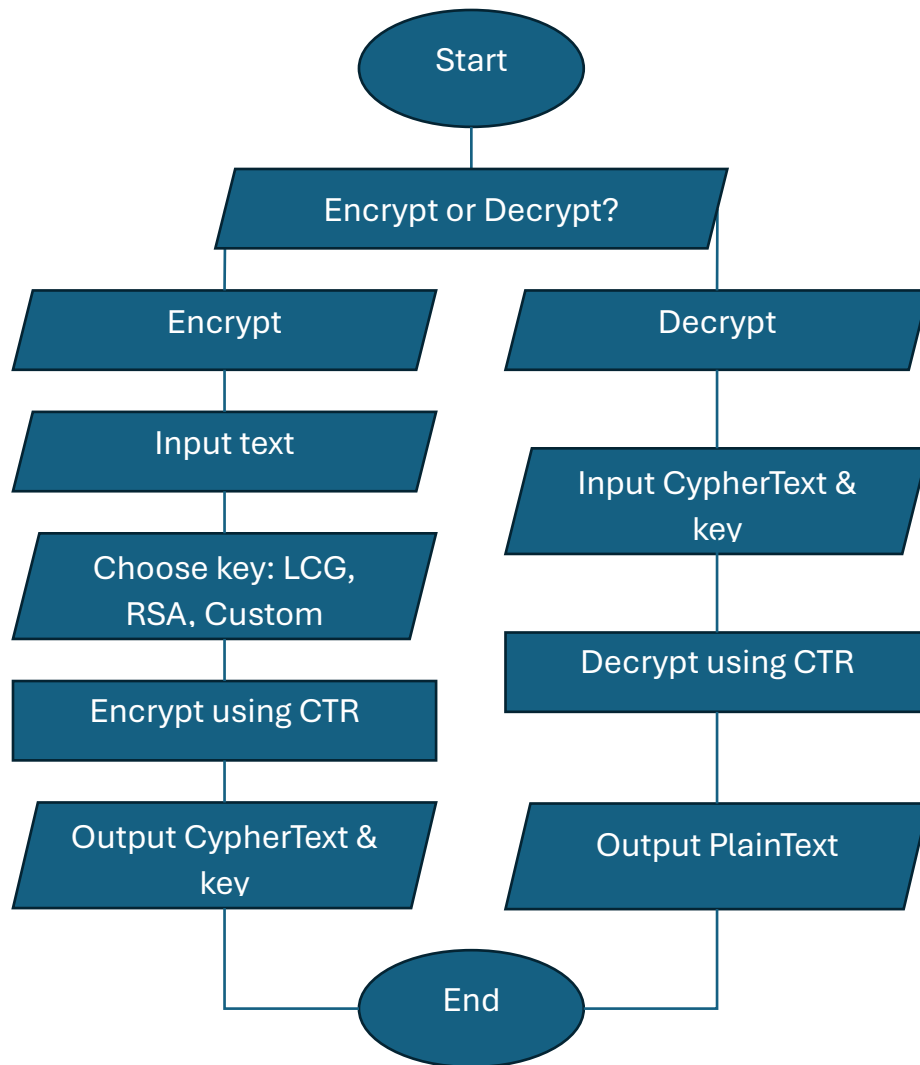
Future Enhancements

- Add support for more encryption algorithms like AES.
- Improve the user interface with animations and better visuals.
- Package the application as a standalone desktop app using tools like PyInstaller or cx_Freeze.

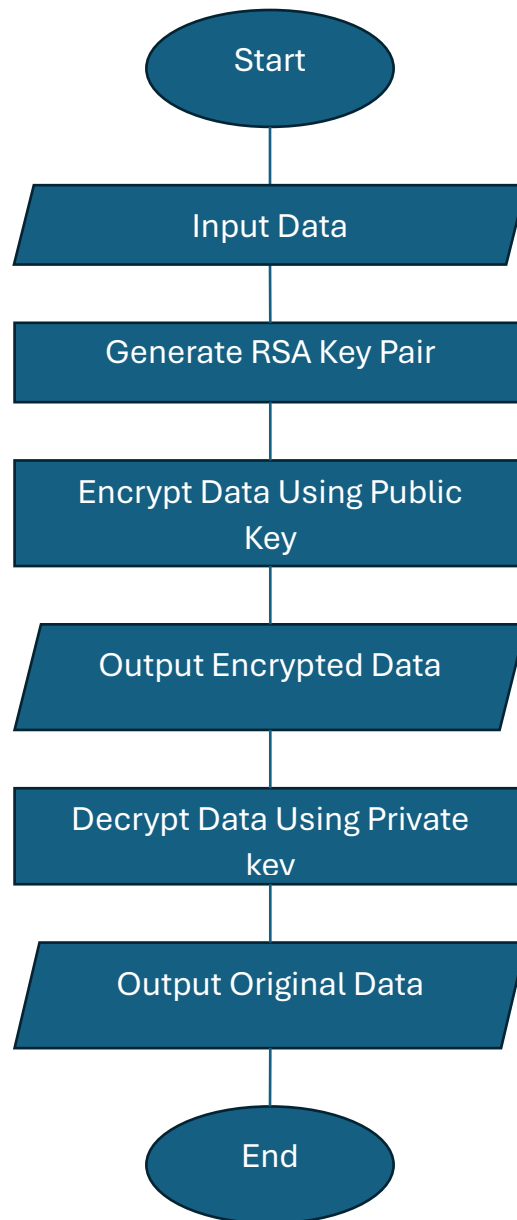
Plain Summary

The project is a web-based encryption and decryption tool built using Python and Flask. It uses CTR Mode for encryption and supports LCG, RSA, and custom keys. The frontend is built with HTML, CSS, and JavaScript, and the application supports multilingual text using UTF-8 encoding. To run the project, install Python, set up the dependencies, and start the Flask server.

Encryption Algorithm in code



RSA Encryption Algorithm



SHA Encryption Algorithm

